Riemann Surfaces

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0.1 Introduction and Motivation

0.1.1 What is a Riemann Surface?

A Riemann surface is a topological space that locally resembles the complex plane $\mathbb{C}$. Just as a globe can be represented by overlapping flat maps in an atlas, a Riemann surface can be covered by overlapping "complex charts" that enable complex analysis at every point.

Formally, a Riemann surface is a one-dimensional complex manifold—a space where we can perform complex analysis even though the space itself may not be a subset of $\mathbb{C}$.

0.1.2 Why Study Riemann Surfaces?

Riemann surfaces are fundamental objects that unify several branches of mathematics:

- **Multi-valued functions:** Functions like $\sqrt{z}$ and $\log z$ are naturally multi-valued on $\mathbb{C}$. Riemann surfaces provide domains where these functions become single-valued and holomorphic.
- **Geometric framework for algebraic curves:** Every smooth algebraic curve over $\mathbb{C}$ can be viewed as a Riemann surface, bridging algebraic geometry and complex analysis.
- **Unification of disciplines:** Riemann surfaces sit at the intersection of topology, complex analysis, and algebraic geometry, providing deep connections between these fields.

0.2 Complex Charts

0.2.1 Definition

Definition 0.2.1. A **complex chart** on a topological space X is a homeomorphism

$$\varphi : U \to V$$

where $U \subseteq X$ is open and $V \subseteq \mathbb{C}$ is open.

Key terminology:

- U is called the **domain** of the chart.
- The chart is **centered** at $p \in U$ if $\varphi(p) = 0$.
- The value $z = \varphi(x)$ serves as a **local complex coordinate** on U.

0.2.2 Examples of Complex Charts

Example 0.2.2 (Standard Charts on $\mathbb{R}^2$)**.** For any open set $U \subseteq \mathbb{R}^2$, define

$$\varphi_U(x, y) = x + iy.$$

This identifies $\mathbb{R}^2$ with $\mathbb{C}$ in the natural way.

Example 0.2.3 (Non-Standard Charts on $\mathbb{R}^2$)**.** For any open $U \subseteq \mathbb{R}^2$, one can define

$$\varphi_U(x, y) = \frac{x}{1 + \sqrt{x^2 + y^2}} + i\frac{y}{1 + \sqrt{x^2 + y^2}}.$$

This maps all of $\mathbb{R}^2$ into the open unit disk.

Example 0.2.4 (Sub-charts)**.** If $\varphi : U \to V$ is a chart and $U_1 \subseteq U$ is open, then the restriction

$$\varphi|_{U_1} : U_1 \to \varphi(U_1)$$

is also a chart, called a **sub-chart**.

Example 0.2.5 (Coordinate Changes)**.** If $\varphi : U \to V$ is a chart and $\psi : V \to W$ is a holomorphic bijection, then the composition

$$\psi \circ \varphi : U \to W$$

is also a chart.

0.3 Compatibility of Charts

0.3.1 The Compatibility Condition

The central question in defining a complex structure is: when should two charts be considered equivalent for purposes of complex analysis?

Definition 0.3.1. Two charts $\varphi_1 : U_1 \to V_1$ and $\varphi_2 : U_2 \to V_2$ are **compatible** if either:

1. $U_1 \cap U_2 = \emptyset$, or
2. The **transition function**

$$T = \varphi_2 \circ \varphi_1^{-1} : \varphi_1(U_1 \cap U_2) \to \varphi_2(U_1 \cap U_2)$$

   is holomorphic.

Remark 0.3.2. Compatibility is symmetric. If $T = \varphi_2 \circ \varphi_1^{-1}$ is holomorphic, then so is $T^{-1} = \varphi_1 \circ \varphi_2^{-1}$, since the inverse of a holomorphic bijection is holomorphic.

0.3.2 Non-Vanishing Derivative of Transition Functions

Lemma 0.3.3. *If T is a transition function between compatible charts, then $T'(w) \neq 0$ for all w in the domain.*

Proof. Let $S = T^{-1}$. Then $S(T(w)) = w$. Differentiating using the chain rule:

$$S'(T(w)) \cdot T'(w) = 1.$$

Therefore $T'(w) \neq 0$. □

Consequence: The power series expansion of any transition function has the form

$$z = T(w) = z_0 + a_1(w - w_0) + a_2(w - w_0)^2 + \cdots$$

with $a_1 \neq 0$.

0.3.3 Examples of Compatible and Incompatible Charts

- **Compatible:** Any two charts from the standard identification $\mathbb{R}^2 \cong \mathbb{C}$ are compatible, since the transition function is the identity on the overlap.
- **Incompatible:** No chart from the standard identification is compatible with any chart from the non-standard example (unless their domains are disjoint). The transition function would involve the non-holomorphic function $\sqrt{x^2 + y^2}$.

0.4 The Riemann Sphere

0.4.1 Construction

The Riemann Sphere is constructed from the unit sphere

$$S^2 = \{(x, y, w) \in \mathbb{R}^3 : x^2 + y^2 + w^2 = 1\}$$

using two stereographic projections:

Chart 1 (from the north pole):

$$\varphi_1 : S^2 \setminus \{(0, 0, 1)\} \to \mathbb{C}, \quad \varphi_1(x, y, w) = \frac{x}{1 - w} + i\frac{y}{1 - w}$$

Chart 2 (from the south pole, with conjugation):

$$\varphi_2 : S^2 \setminus \{(0, 0, -1)\} \to \mathbb{C}, \quad \varphi_2(x, y, w) = \frac{x}{1 + w} - i\frac{y}{1 + w}$$

0.4.2 Transition Function

On the overlap $S^2 \setminus \{(0, 0, \pm 1)\}$, the transition function is:

$$\varphi_2 \circ \varphi_1^{-1}(z) = \frac{1}{z}$$

This is holomorphic on $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$, confirming that the charts are compatible.

0.4.3 Properties of the Riemann Sphere

- One chart covers $\mathbb{C}$ with coordinate z.
- The other covers a neighborhood of ∞ with coordinate $1/z$.
- The Riemann Sphere is compact.
- Often denoted $\mathbb{C}_\infty$ or $\mathbb{C} \cup \{\infty\}$.

0.5 Complex Atlases and Complex Structures

0.5.1 Complex Atlas

Definition 0.5.1. A **complex atlas** $\mathcal{A}$ on X is a collection $\mathcal{A} = \{\varphi_\alpha : U_\alpha \to V_\alpha\}$ of pairwise compatible charts whose domains cover X:

$$X = \bigcup_\alpha U_\alpha$$

Example 0.5.2.
- The collection of all standard charts on $\mathbb{R}^2$ forms an atlas.
- The two stereographic charts φ_1, φ_2 form an atlas on S^2.

0.5.2 Induced Atlas on Open Subsets

If $\mathcal{A} = \{\varphi_\alpha : U_\alpha \to V_\alpha\}$ is an atlas on X and $Y \subseteq X$ is open, then

$$\mathcal{A}_Y = \{\varphi_\alpha|_{Y \cap U_\alpha}\}$$

is an atlas on Y.

0.5.3 Equivalent Atlases

Definition 0.5.3. Two atlases $\mathcal{A}$ and $\mathcal{B}$ are **equivalent** if every chart of one is compatible with every chart of the other.

Key Fact: Two atlases are equivalent if and only if their union is also an atlas.

0.5.4 Complex Structure

Definition 0.5.4. A **complex structure** on X is:

- A maximal complex atlas, or equivalently,
- An equivalence class of atlases.

Practical Approach: To define a complex structure, one need only provide any atlas—it uniquely determines a maximal one.

0.6 Definition of a Riemann Surface

0.6.1 Formal Definition

Definition 0.6.1. A **Riemann surface** is a second countable, connected, Hausdorff topological space X together with a complex structure.

The topological conditions ensure:

- **Hausdorff:** Any two distinct points have disjoint neighborhoods.
- **Second countable:** The topology has a countable basis.

0.6.2 Fundamental Examples

- **The Complex Plane:** $X = \mathbb{C}$ with the standard atlas.
- **The Riemann Sphere:** $X = S^2$ with the two-chart atlas from stereographic projection, often written as $\mathbb{C}_\infty$ or $\mathbb{C} \cup \{\infty\}$.
- **Open Subsets:** Any connected open subset of a Riemann surface is itself a Riemann surface.

0.7 Riemann Surfaces as Real Manifolds

0.7.1 Relationship to Smooth Manifolds

A C^∞ real manifold is defined by replacing "holomorphic" with "C^∞ diffeomorphism" in all definitions.

Key Observation: Since holomorphic maps are C^∞ (as functions of real variables), every Riemann surface is automatically a 2-dimensional C^∞ real manifold.

0.7.2 Orientability

Proposition 0.7.1. *Every Riemann surface is an orientable, path-connected, 2-dimensional C^∞ real manifold.*

Why orientable? Holomorphic maps preserve orientation—they preserve angles, including the sense of rotation.

0.8 The Genus of a Compact Riemann Surface

0.8.1 Classification of Compact Surfaces

Theorem 0.8.1 (Classification Theorem). *Every compact orientable 2-manifold is homeomorphic to a "g-holed torus" for a unique integer $g \geq 0$.*

Genus g	Topological Description
0	Sphere
1	Torus ($S^1 \times S^1$)
2	Two-holed torus
g	Surface with g handles

0.8.2 Topological Genus

Proposition 0.8.2. *Every compact Riemann surface is diffeomorphic to a g-holed torus for some unique $g \geq 0$. This g is called the* ***topological genus****.*

Example 0.8.3. The Riemann Sphere has genus 0.

0.9 Higher-Dimensional Generalization: Complex Manifolds

0.9.1 Definition

Definition 0.9.1. An n-dimensional **complex manifold** is defined analogously, using charts $\varphi : U \to V \subseteq \mathbb{C}^n$ with transition functions holomorphic in each variable.

A Riemann surface is thus a 1-dimensional complex manifold.

0.9.2 The Projective Line $\mathbb{P}^1$

The projective line is defined as:

$$\mathbb{CP}^1 = \mathbb{P}^1 = \{\text{1-dimensional subspaces of } \mathbb{C}^2\}$$

A point is written $[z : w]$ where $(z, w) \neq (0, 0)$, with the equivalence relation:

$$[z : w] = [\lambda z : \lambda w] \quad \text{for all } \lambda \in \mathbb{C}^*$$

0.9.3 Atlas on $\mathbb{P}^1$

Chart	Domain	Map
φ_0	$U_0 = \{[z : w] : z \neq 0\}$	$\varphi_0[z : w] = w/z$
φ_1	$U_1 = \{[z : w] : w \neq 0\}$	$\varphi_1[z : w] = z/w$

Transition Function: On $U_0 \cap U_1$:

$$\varphi_1 \circ \varphi_0^{-1}(s) = 1/s$$

Verification:

- $U_0 \cup U_1 = \mathbb{P}^1$ ✓
- $\varphi_i(U_0 \cap U_1) = \mathbb{C}^*$ is open ✓
- Transition function is holomorphic ✓
- Connected: U_0, U_1 connected with nonempty intersection ✓

0.10 Complex Tori

0.10.1 Setup and Construction

Fix $\omega_1, \omega_2 \in \mathbb{C}$ linearly independent over $\mathbb{R}$. Define the **lattice**:

$$L = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 = \{m_1\omega_1 + m_2\omega_2 : m_1, m_2 \in \mathbb{Z}\}$$

The quotient space is:

$$X = \mathbb{C}/L$$

with projection $\pi : \mathbb{C} \to X$.

Topology: $U \subseteq X$ is open if and only if $\pi^{-1}(U)$ is open in $\mathbb{C}$.

0.10.2 Key Properties

- π is an open map: If V is open in $\mathbb{C}$, then $\pi^{-1}(\pi(V)) = \bigcup_{\omega \in L}(\omega + V)$ is open.
- X is connected: It is the image of connected $\mathbb{C}$.
- X is compact: It is the image of a compact fundamental parallelogram.

0.10.3 Constructing Charts

Choose $\epsilon > 0$ such that $|\omega| > 2\epsilon$ for all nonzero $\omega \in L$.

For each $z_0 \in \mathbb{C}$, let $D_{z_0} = D(z_0, \epsilon)$. Then:

- $\pi|_{D_{z_0}} : D_{z_0} \to \pi(D_{z_0})$ is a homeomorphism.
- Define $\varphi_{z_0} : \pi(D_{z_0}) \to D_{z_0}$ as its inverse.

Compatibility Check: For two charts φ_{z_1} and φ_{z_2} with overlapping domains:

$$T(z) = \varphi_{z_2}(\varphi_{z_1}^{-1}(z)) = z + \omega(z)$$

where $\omega(z) \in L$ is locally constant. Hence T is locally of the form $z \mapsto z + \omega$, which is holomorphic.

0.10.4 Main Theorem

Theorem 0.10.1. *$\mathbb{C}/L$ is a compact Riemann surface of genus 1 (topologically a torus $S^1 \times S^1$).*

Proof. $\mathbb{C}/\Lambda$ inherits a holomorphic structure from $\mathbb{C}$ because the translation group $\Lambda \subset \mathrm{Aut}(\mathbb{C})$ acts freely and properly discontinuously. It is compact because it is the continuous image of a closed, bounded fundamental parallelogram $P \subset \mathbb{C}$ under the quotient map $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$. Topologically, $\mathbb{C}/\Lambda \cong \mathbb{R}^2/\mathbb{Z}^2 \cong S^1 \times S^1$, which yields an Euler characteristic $\chi = 0$, implying genus $g = 1$ via the relation $\chi = 2 - 2g$. □

0.11 Graphs of Holomorphic Functions

0.11.1 Construction

Let $V \subseteq \mathbb{C}$ be connected and open, and let $g : V \to \mathbb{C}$ be holomorphic. Define:

$$X = \{(z, g(z)) : z \in V\} \subseteq \mathbb{C}^2$$

Chart: The projection $\pi : X \to V$ given by $\pi(z, w) = z$ is a homeomorphism. Its inverse $\pi^{-1}(z) = (z, g(z))$ gives a single chart covering all of X.

0.11.2 Riemann Surface Structure

Theorem 0.11.1. *X is a Riemann surface (with a one-chart atlas).*

Proof. Let X be a topological space equipped with an atlas $\mathcal{A} = \{(X, \phi)\}$, where $\phi : X \to U \subseteq \mathbb{C}$ is a homeomorphism. For X to be a Riemann surface, the transition maps between overlapping charts must be holomorphic. In the case of a single-chart atlas, the only transition map is the identity $\psi = \phi \circ \phi^{-1} : U \to U$, defined by $\psi(z) = z$. Since the identity map is holomorphic on U, the atlas $\mathcal{A}$ is holomorphically compatible. Thus, X is a complex manifold of dimension 1, which by definition is a Riemann surface. □

0.12 The Implicit Function Theorem

0.12.1 Statement

Theorem 0.12.1 (Implicit Function Theorem)**.** *Let $f(z, w) \in \mathbb{C}[z, w]$ and $X = \{(z, w) : f(z, w) = 0\}$. If $p = (z_0, w_0) \in X$ and $\frac{\partial f}{\partial w}(p) \neq 0$, then near p, X is a graph $w = g(z)$ for some holomorphic g.*

Moreover:

$$g' = -\frac{\partial f/\partial z}{\partial f/\partial w}$$

0.12.2 Proof

Since $f \in \mathbb{C}[z, w]$ is holomorphic and $\frac{\partial f}{\partial w}(p) \neq 0$, the Complex Implicit Function Theorem guarantees the existence of an open neighborhood $U \ni z_0$ and a unique holomorphic function $g : U \to \mathbb{C}$ such that $f(z, g(z)) = 0$ and $g(z_0) = w_0$.

To find g', differentiate the identity $f(z, g(z)) = 0$ with respect to z using the chain rule:

$$\frac{\partial f}{\partial z}(z, g(z)) + \frac{\partial f}{\partial w}(z, g(z)) \cdot g'(z) = 0$$

Rearranging:

$$g'(z) = -\frac{\partial f/\partial z}{\partial f/\partial w}\bigg|_{(z,g(z))}$$

0.13 Smooth Affine Plane Curves

0.13.1 Definition

Definition 0.13.1. An **affine plane curve** is $X = \{(z, w) \in \mathbb{C}^2 : f(z, w) = 0\}$.

The curve is **smooth** (or nonsingular) at $p \in X$ if:

$$\left(\frac{\partial f}{\partial z}(p), \frac{\partial f}{\partial w}(p)\right) \neq (0, 0)$$

The curve is **smooth** if it is smooth at every point.

0.13.2 Constructing Charts

At each $p \in X$:

- If $\frac{\partial f}{\partial w}(p) \neq 0$: use projection onto the z-axis.
- If $\frac{\partial f}{\partial z}(p) \neq 0$: use projection onto the w-axis.

0.13.3 Connectedness

Theorem 0.13.2 (Connectedness)**.** *If $f(z,w)$ is irreducible (cannot be factored as a product of non-constant polynomials), then X is connected.*

Proof. Suppose X is not connected. Then $X = X_1 \cup X_2$ for two non-empty, disjoint, closed sets X_1 and X_2. By Hilbert's Nullstellensatz, every algebraic component of X corresponds to a factor of f.

If X were disconnected, it would imply that X is the union of at least two distinct algebraic varieties $V(f_1)$ and $V(f_2)$. This would mean f can be factored as $f = f_1^a \cdot f_2^b$ for some non-constant polynomials f_1 and f_2.

This contradicts the assumption that f is irreducible in the ring $\mathbb{C}[z,w]$. Since the set of regular points $X \setminus \mathrm{Sing}(X)$ is a connected Riemann surface and $\mathrm{Sing}(X)$ is a finite set of points, the total space X must be connected. □

0.13.4 Main Result

Theorem 0.13.3. *Every smooth irreducible affine plane curve is a Riemann surface.*

Proof. Let $f \in \mathbb{C}[z,w]$ be an irreducible polynomial. The smoothness of $X = V(f)$ implies that for every $p \in X$, $df_p \neq 0$. By the Complex Implicit Function Theorem, for each $p \in X$, there exists a neighborhood $U \subset X$ and a holomorphic chart $\phi : U \to \mathbb{C}$ (projecting onto either the z or w coordinate). The transition maps between these charts are holomorphic wherever they are defined.

Since f is irreducible, X is connected in the classical topology. A connected, one-dimensional complex manifold is a Riemann surface; thus, X is a Riemann surface. □

Remark 0.13.4. Affine plane curves are never compact (unbounded in $\mathbb{C}^2$).

0.14 Hyperelliptic Curves

0.14.1 Definition

Let $h(z)$ be a polynomial that is not a perfect square. Consider:

$$f(z,w) = w^2 - h(z)$$

0.14.2 Properties

- f is irreducible if and only if h is not a perfect square.
- f is nonsingular if and only if h has distinct roots.

0.14.3 Verification of Smoothness

Computing the partial derivatives:

$$\frac{\partial f}{\partial z} = -h'(z), \qquad \frac{\partial f}{\partial w} = 2w$$

At a singular point: $w = 0$ and $h'(z) = 0$ and $h(z) = 0$.
This means z is a repeated root of h. So if h has distinct roots, X is smooth.

0.15 Holomorphic Functions on Riemann Surfaces

0.15.1 Definition

Definition 0.15.1. A function $f : W \to \mathbb{C}$ on an open set W of a Riemann surface X is **holomorphic** at $p \in W$ if for some (equivalently, every) chart $\varphi : U \to V$ with $p \in U$:

$$f \circ \varphi^{-1} : V \to \mathbb{C}$$

is holomorphic at $\varphi(p)$.

0.15.2 Chart Independence

Lemma 0.15.2. *The definition is independent of the choice of chart because:*

$$f \circ \varphi_2^{-1} = (f \circ \varphi_1^{-1}) \circ (\varphi_1 \circ \varphi_2^{-1})$$

is a composition of holomorphic functions.

0.15.3 Notation

$$\mathcal{O}(W) = \mathcal{O}_X(W) = \{f : W \to \mathbb{C} : f \text{ is holomorphic}\}$$

This forms a $\mathbb{C}$-algebra.

0.15.4 Examples

- **Charts as holomorphic functions:** Any chart $\varphi : U \to V$, viewed as a function $U \to \mathbb{C}$, is holomorphic.
- **Functions on $\mathbb{C}_\infty$ near ∞:** f is holomorphic at ∞ if and only if $f(1/z)$ is holomorphic at $z = 0$. For a rational function $p(z)/q(z)$: holomorphic at ∞ if and only if $\deg(p) \le \deg(q)$.
- **Functions on $\mathbb{P}^1$:** If $p(z, w)$ and $q(z, w)$ are homogeneous polynomials of the same degree with $q(z_0, w_0) \neq 0$, then

  $$f([z : w]) = \frac{p(z, w)}{q(z, w)}$$

  is holomorphic near $[z_0 : w_0]$.
- **Functions on Tori:** $f : \mathbb{C}/L \to \mathbb{C}$ is holomorphic if and only if $f \circ \pi : \mathbb{C} \to \mathbb{C}$ is holomorphic.
- **Functions on Affine Curves:** On a smooth affine curve $X = \{f(z, w) = 0\}$:
  - Coordinate projections are holomorphic.
  - Any polynomial $g(z, w)$ restricted to X is holomorphic.

0.16 Types of Singularities

0.16.1 Classification

For f holomorphic in a punctured neighborhood of p:

Type	**Condition**	**Behavior of $\lvert f\rvert$**
Removable	$f \circ \varphi^{-1}$ has removable singularity	Bounded near p
Pole	$f \circ \varphi^{-1}$ has a pole	$\to \infty$ as $x \to p$
Essential	$f \circ \varphi^{-1}$ has essential singularity	No limit

0.16.2 Detailed Descriptions

- **Removable:** By Riemann's Theorem on Removable Singularities, if f is bounded in a punctured disk $D \setminus \{p\}$, the singularity can be "filled" by defining $f(p)$ such that f becomes holomorphic at p.

- **Pole:** Near a pole of order m, $1/f$ has a removable singularity at p. After extending $1/f$ holomorphically, we have $(1/f)(p) = 0$, and $f(z)$ can be expressed locally as $f(z) = (z-p)^{-m}h(z)$ where $h(p) \neq 0$.

- **Essential:** By the Casorati-Weierstrass Theorem, if p is an essential singularity, the image $f(D \setminus \{p\})$ is dense in $\mathbb{C}$. For any $w \in \mathbb{C}$ and any $\epsilon > 0$, there exists z near p such that $|f(z) - w| < \epsilon$.

0.17 Meromorphic Functions

0.17.1 Definition

Definition 0.17.1. f is **meromorphic** at p if it is holomorphic, has a removable singularity, or has a pole at p.

Notation:

$$\mathcal{M}(W) = \{f : W \to \mathbb{C} : f \text{ is meromorphic}\}$$

0.17.2 Examples

- **Rational functions:** Any rational function is meromorphic on all of $\mathbb{C}_\infty$.

- **Elliptic functions:** A meromorphic L-periodic function on $\mathbb{C}$ is called an **elliptic function**. These correspond bijectively to meromorphic functions on the torus $\mathbb{C}/L$.

0.18 Laurent Series and Order

0.18.1 Laurent Series

For f holomorphic in a punctured neighborhood of p, with chart φ giving coordinate z:

$$f(\varphi^{-1}(z)) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$$

0.18.2 Characterization of Singularities

Singularity Type	**Laurent Series Condition**
Removable	No negative terms
Pole	Finitely many negative terms
Essential	Infinitely many negative terms

0.18.3 Order of a Meromorphic Function

Definition 0.18.1.

$$\text{ord}_p(f) = \min\{n : c_n \neq 0\}$$

Chart Independence: If $z = T(w) = z_0 + a_1(w - w_0) + \ldots$ with $a_1 \neq 0$, then substituting preserves the minimum exponent.

0.18.4 Interpretation of Order

Condition	Meaning
$\text{ord}_p(f) \geq 0$	f holomorphic at p
$\text{ord}_p(f) > 0$	f has a zero at p
$\text{ord}_p(f) < 0$	f has a pole at p
$\text{ord}_p(f) = 0$	$f(p) \neq 0$ and f holomorphic

0.18.5 Algebraic Properties

Lemma 0.18.2. *For nonzero meromorphic f, g:*

$$\text{ord}_p(fg) = \text{ord}_p(f) + \text{ord}_p(g) \tag{1}$$
$$\text{ord}_p(f/g) = \text{ord}_p(f) - \text{ord}_p(g) \tag{2}$$
$$\text{ord}_p(f+g) \geq \min\{\text{ord}_p(f), \text{ord}_p(g)\} \tag{3}$$

0.18.6 Example: Rational Functions on $\mathbb{C}_\infty$

Let $f(z) = c\prod_i (z - \lambda_i)^{e_i}$ be a rational function.

Then:

- $\text{ord}_{\lambda_i}(f) = e_i$
- $\text{ord}_\infty(f) = -\sum_i e_i$
- $\text{ord}_x(f) = 0$ for all other x

Sum Formula:

$$\sum_{x \in \mathbb{C}_\infty} \text{ord}_x(f) = 0$$

This fundamental property generalizes to all compact Riemann surfaces.

0.19 C^∞ and Harmonic Functions

0.19.1 Definitions

Definition 0.19.1. C^∞ **Functions:** f is C^∞ at p if $f \circ \varphi^{-1}$ has continuous partial derivatives of all orders.

Definition 0.19.2. Harmonic Functions: A real-valued C^∞ function $h(x, y)$ is **harmonic** if:

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = 0$$

0.19.2 Well-Definedness

Lemma 0.19.3. *Harmonicity is well-defined on a Riemann surface (independent of chart choice).*

Fact: Real and imaginary parts of holomorphic functions are harmonic.

0.20 Fundamental Theorems

0.20.1 Discreteness of Zeros and Poles

Theorem 0.20.1. *For f meromorphic and not identically zero on a connected open W, the zeros and poles of f form a discrete subset of W.*

Proof. **Zeros:** If the set of zeros $Z = \{z \in W : f(z) = 0\}$ has a limit point in W, the Identity Theorem implies $f \equiv 0$ on W. Since $f \not\equiv 0$, all zeros must be isolated, making Z a discrete subset.

Poles: By definition, each pole p is an isolated singularity. The set of poles P corresponds to the zeros of the holomorphic function $h = 1/f$ in local neighborhoods. Applying the Identity Theorem to h, we find that its zeros (and thus the poles of f) are isolated.

Therefore, the set $Z \cup P$ of zeros and poles is a discrete subset of W. □

Corollary 0.20.2. *On a compact Riemann surface, any non-zero meromorphic function has finitely many zeros and poles.*

0.20.2 Identity Theorem

Theorem 0.20.3. *If $f = g$ on a set with a limit point in a connected W, then $f = g$ on all of W.*

Proof. Define $h(z) = f(z) - g(z)$. Then h is holomorphic on W. We wish to show that if the set of zeros $Z_h = \{z \in W : h(z) = 0\}$ has a limit point $a \in W$, then $h \equiv 0$ on W.

Local Behavior: Since h is holomorphic at a, it can be represented by a Taylor series in some disk $D(a, R) \subset W$:

$$h(z) = \sum_{n=0}^{\infty} c_n (z - a)^n$$

Suppose h is not identically zero on $D(a, R)$. Then there exists a smallest integer k such that $c_k \neq 0$. We can factor:

$$h(z) = (z - a)^k \phi(z)$$

where $\phi(z)$ is holomorphic and $\phi(a) = c_k \neq 0$. By continuity of ϕ, there exists a neighborhood of a where $\phi(z) \neq 0$. In this neighborhood, $h(z) = 0$ only at the isolated point $z = a$. This contradicts the assumption that a is a limit point of zeros. Thus, $c_n = 0$ for all n, and $h \equiv 0$ on $D(a, R)$.

Global Argument: Let $U = \{z \in W : z \text{ is a limit point of } Z_h\}$.

- U is non-empty by hypothesis.
- U is open: If $z_0 \in U$, the argument above shows $h \equiv 0$ in a neighborhood of z_0.
- U is closed in W: If $z_n \in U$ and $z_n \to z \in W$, then by continuity $h(z) = 0$, and z is a limit point.

Since W is connected and U is non-empty, open, and closed, $U = W$. Therefore $h \equiv 0$ on W, implying $f = g$. □

0.20.3 Maximum Modulus Theorem

Theorem 0.20.4. *If f is holomorphic on connected W and $|f|$ achieves its maximum at an interior point, then f is constant.*

Proof. Suppose $|f(z_0)| = M$ is a local maximum. Choose a small disk $D(z_0, r) \subset W$ such that $|f(z)| \leq M$ for all $z \in D(z_0, r)$. By the Cauchy Integral Formula, we have the Mean Value Property:

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta})\, d\theta \quad \text{for } 0 < \rho < r$$

Taking the absolute value:

$$M = |f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + \rho e^{i\theta})|\, d\theta$$

Since $|f(z_0 + \rho e^{i\theta})| \leq M$, the only way the average can equal M is if $|f(z_0 + \rho e^{i\theta})| = M$ for all θ.

Thus $|f(z)|$ is constant on $D(z_0, r)$. A holomorphic function with constant modulus is itself constant. Since W is connected and f is constant on an open subset, the Identity Theorem implies f is constant on all of W. □

0.20.4 The Fundamental Theorem for Compact Surfaces

Theorem 0.20.5. *If X is a compact Riemann surface and f is holomorphic on all of X, then f is constant.*

Proof. Since X is compact, $|f|$ achieves its maximum at some point $p \in X$. By the Maximum Modulus Theorem, f is constant. □

Interpretation: There are no non-constant holomorphic functions on compact Riemann surfaces—they can only have meromorphic functions with poles.

0.20.5 Maximum Principle for Harmonic Functions

Theorem 0.20.6. *Any harmonic function on a compact Riemann surface is constant.*

Proof. Let u be a harmonic function on X. Since X is compact and u is continuous, u must attain its maximum value M at some point $p \in X$ by the Extreme Value Theorem.

Let (U, ϕ) be a local holomorphic chart around p. The function $u \circ \phi^{-1}$ is harmonic on an open subset of $\mathbb{R}^2$ and attains a local maximum at $\phi(p)$. By the Strong Maximum Principle for harmonic functions, $u \circ \phi^{-1}$ must be constant on that neighborhood, implying u is constant on U.

The set $S = \{x \in X : u(x) = M\}$ is non-empty (contains p) and closed (by continuity). The local argument shows S is also open. Since X is connected, $S = X$. Thus $u \equiv M$ everywhere. □